

CASER: Cost-Aware Selective Enhancement for Conversational Retrieval

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Abstract—In multi-turn question answering, rewriting the current query according to the dialogue history can effectively mitigate coreference, ellipsis, and semantic incompleteness, while reranking retrieved candidate documents with a cross-encoder can further improve ranking quality. Enhancement pipelines consisting of query rewriting and document reranking have been widely used to improve conversational retrieval. However, these methods usually require additional model inference, which introduces extra latency and computational cost. In practical applications, different queries and candidate document sets do not benefit equally from enhancement operations, and some samples obtain only limited marginal gains after the full enhancement pipeline is applied. Therefore, under budget-constrained scenarios, it is necessary to selectively perform query rewriting and document reranking according to the potential gain of each sample. To address this problem, this paper proposes CASER, a cost-aware selective enhancement framework for conversational retrieval. CASER formulates query rewriting and document reranking under a unified budget constraint, estimates the potential retrieval-quality improvement of different enhancement operations with gain predictors, and selectively triggers the corresponding operations by considering their costs. Experiments on the QReCC dataset show that CASER can substantially reduce enhancement cost while maintaining retrieval effectiveness close to the full enhancement pipeline, demonstrating the effectiveness of cost-aware selective enhancement for conversational retrieval.

Index Terms—conversational retrieval, query rewriting, document reranking, cost-aware retrieval, gain prediction, budget constraint

I. INTRODUCTION

Open-domain question answering and retrieval-augmented generation systems increasingly adopt conversational interaction. Unlike single-turn queries, conversational queries often depend on previous turns through coreference, ellipsis, or contextual continuation. Expressions such as “he” or “this movie” may be ambiguous without dialogue history, leading to irrelevant retrieval results. Query rewriting addresses this problem by transforming context-dependent questions into self-contained queries, improving subsequent retrieval accuracy [1], [2].

Candidate document reranking further improves retrieval quality. An initial retriever efficiently returns candidates from a large corpus, after which cross-encoders model query-document interactions to produce more accurate relevance estimates [3], [4]. However, scoring these candidate pairs re-

quires additional computation and can contribute substantially to latency when the candidate set is large.

Applying both operations to every query may be impractical under latency, computing-resource, or API-budget constraints. Moreover, their benefits vary across samples. Here, a sample consists of the current conversational turn q_i and its preceding dialogue history H_i , rather than an entire conversation. A clear query may benefit little from rewriting, while an already satisfactory candidate ranking may gain little from reranking. These differences motivate allocating limited enhancement resources across samples and operations according to their potential benefits, rather than applying the full pipeline uniformly.

To address this problem, we propose CASER, a cost-aware selective enhancement framework for conversational retrieval. CASER treats query rewriting and document reranking as optional operations with different costs and benefits, using gain predictors to guide their selection under a shared budget. Its sequential strategy follows the retrieval pipeline: rewriting is selected before retrieval using the current query and dialogue history, while reranking is selected afterward using the final query and retrieved candidate statistics. This design coordinates both operations while respecting the information available at each decision stage. The main contributions of this paper are as follows:

- We formulate query rewriting and document reranking as optional enhancement operations under a unified budget, expressed relative to full enhancement, and study their allocation through the cost-quality trade-off.
- We design CASER, a sequential selection framework that estimates expected positive retrieval gains and combines them with operation costs, using the information available before and after retrieval.
- We evaluate CASER on QReCC against full enhancement and alternative budget allocation strategies, complemented by component ablations and single-module operating points. Results show that CASER maintains retrieval effectiveness close to full enhancement at lower modeled enhancement cost.

II. RELATED WORK

A. Conversational Retrieval and Query Rewriting

Conversational retrieval uses dialogue history to identify the user’s current information need and retrieve relevant documents. Query rewriting resolves contextual dependencies by

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converting follow-up questions into standalone queries. The CANARD dataset [1] defines the question-in-context rewriting task using self-contained questions. QReCC [2] extends this setting to open-domain conversational question answering, providing original questions, human rewrites, answers, and passage retrieval annotations for evaluating rewriting, retrieval, and reading comprehension. TREC CAsT further advances conversational retrieval, with systems combining query rewriting, expansion, dense retrieval, and neural reranking [5].

Beyond generating complete rewrites, other methods select historical information or learn context-aware representations. QuReTeC [6] classifies historical terms to determine whether they should be added to the current query. ConvDR [7] uses teacher–student training to learn contextualized query representations for dense retrieval. More recent generative approaches produce informative LLM rewrites or combine query reformulation with potential answer generation [8], [9]. These methods improve query understanding through dialogue context. CASER addresses a complementary question: whether the expected benefit of rewriting justifies its cost when rewriting and reranking share a limited enhancement budget.

B. Two-Stage Retrieval and Candidate Document Reranking

Open-domain question answering and conversational retrieval systems commonly follow a retrieve-and-rerank pipeline. Initial retrieval obtains candidates from a large corpus using sparse term matching or dense representations. BM25 is a standard sparse retrieval method [10], while DPR separately encodes questions and passages with dual encoders and retrieves by vector similarity [11]. Finer-grained matching models then rerank these candidates to improve the final ranking.

Pretrained language models provide a foundation for cross-encoder reranking [4]. BERT rerankers jointly encode concatenated query–passage pairs and estimate relevance through deep interaction modeling [3]. monoT5 instead formulates ranking as a sequence-to-sequence generation task [12]. Although cross-encoders provide fine-grained relevance estimates, scoring each candidate pair incurs computation that grows with candidate count. ColBERT reduces online overhead through late interaction while retaining fine-grained matching [13].

Improving ranking quality or reducing model computation addresses the effectiveness and efficiency of reranking, but does not determine whether every query benefits enough to justify its cost. When retrieved candidates are already well ranked, further reranking may offer limited improvement. This motivates treating reranking as an optional enhancement, selected according to its expected gain and cost under the available budget.

C. Cost-Aware Retrieval and Selective Enhancement

Practical retrieval systems must balance effectiveness, latency, and computational resources. Cascaded ranking filters candidates through inexpensive models before applying costly models to a smaller set [14], [15]. For neural document ranking, Intra-Document Cascading selects passages within

each document before expensive scoring, reducing reranking latency [16]. FastBERT adapts inference depth across samples to reduce redundant computation [17]. ADACUR addresses expensive cross-encoder search through adaptive multi-round CUR decomposition [18].

These approaches improve efficiency through candidate pruning, model acceleration, or interaction computation. CASER addresses allocation between two optional operations at different pipeline stages: rewriting changes the query used for retrieval, while reranking operates on the resulting candidates. Their dependency makes shared budget allocation distinct from optimizing either module alone. Our experiments hold retrieval components fixed to isolate allocation effects. Comparisons with the above methods would require adaptation to the same pipeline and cost accounting; such experiments are not reported here.

III. METHOD

CASER selects optional rewriting and reranking using predicted positive gains and operation costs under a shared budget. Full enhancement provides a cost and quality reference; the objective is to maximize retrieval quality within the available allowance.

A. Problem Definition

Given the current query q_i and its preceding dialogue history H_i , the system retrieves relevant documents from corpus $\mathcal{D}$. The final retrieval query is $q_i^* = q_i$ if rewriting is skipped and $q_i^* = q_i^w$ otherwise. After retrieval, an optional cross-encoder reranks the candidates by re-estimating their relevance to the final query.

For sample i , binary decisions $a_i^w, a_i^r \in \{0, 1\}$ indicate whether rewriting and reranking are performed. A sample represents one current conversational turn with its history, so it permits at most one rewrite and one reranking of the candidate set; a conversation contributes multiple samples. The corresponding costs c_w and c_r count only additional enhancement relative to Direct, excluding base retrieval.

For N samples, full enhancement costs $C_{\text{full}} = N(c_w + c_r)$. A budget ratio $\rho \in [0, 1]$ gives an allowance $B_\rho = \rho N(c_w + c_r)$. Conversely, a fixed enhancement budget B over a known query set can be expressed as $\rho = B/[N(c_w + c_r)]$. Ratios facilitate comparison across query-set sizes under the same cost model without excluding fixed-budget deployment. The ideal objective is to maximize retrieval quality within this allowance:

$$\begin{aligned} \max \quad & \sum_{i=1}^N Q_i(a_i^w, a_i^r) \\ \text{s.t.} \quad & \frac{\sum_{i=1}^N (c_w a_i^w + c_r a_i^r)}{N(c_w + c_r)} \leq \rho. \end{aligned} \tag{1}$$

where $Q_i(a_i^w, a_i^r)$ denotes retrieval quality under the selected operations, measured primarily by nDCG@10. Since true gains are unknown before execution, CASER uses predicted positive gains and operation costs to guide selection.

B. Framework Pipeline

CASER follows the execution order of conversational retrieval. It first predicts the expected positive gain of rewriting from the current query and dialogue history, then compares the gain per unit cost with the threshold $\lambda(\rho)$. After retrieval with the selected query q_i^* , it predicts reranking gain using the final query and candidate statistics, applying the same gain-per-cost threshold.

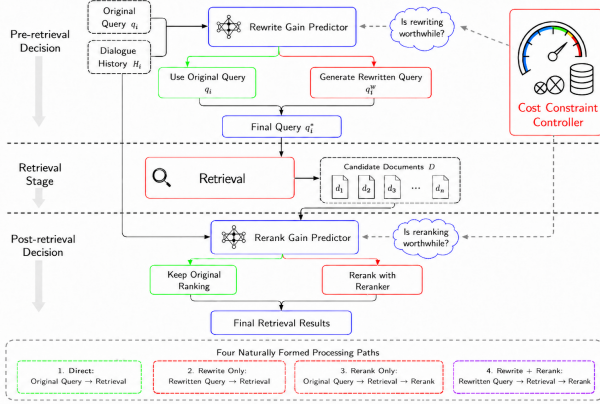


Fig. 1. Overview of CASER: pre-retrieval rewriting selection and post-retrieval reranking selection under a shared cost constraint.

As shown in Fig. 1, the two decisions produce four processing paths: direct retrieval, rewrite only, rerank only, and rewrite plus rerank. Each turn follows one path, with reranking selection conditioned on the candidates returned by the chosen retrieval query.

Sequential prediction respects information availability: rewriting uses pre-retrieval inputs, while reranking uses the selected path’s candidate statistics, reflecting changes induced by rewriting.

C. Gain Label Construction and Expected Positive Gain Prediction

We construct supervised gain labels from offline retrieval and reranking results. For rewriting, retrieval is performed with both the original and rewritten queries, and their nDCG@10 difference defines the gain:

$$g_i^w = Q_i^w - Q_i^d, \quad (2)$$

where Q_i^d and Q_i^w denote original-query and rewritten-query retrieval quality. Reranking labels are constructed separately for these two paths: $g_i^{r,d} = Q_i^{d,r} - Q_i^d$ and $g_i^{r,w} = Q_i^{w,r} - Q_i^w$, where $Q_i^{d,r}$ and $Q_i^{w,r}$ denote quality after reranking. At evaluation, the system uses the Direct-path prediction when rewriting is skipped and the Rewrite-path prediction otherwise, keeping the gain estimate consistent with the actual retrieval path.

We train separate rewriting and reranking predictors, with the latter jointly trained on both paths and their corresponding candidate statistics. Because enhancement gains include positive, zero, and negative outcomes, each predictor uses

two stages: the first estimates the probability of positive gain, $\hat{p}_i = P(g_i > 0 | x_i)$, and the second estimates its conditional magnitude, $\hat{g}_i^+ = E(g_i | g_i > 0, x_i)$. Their product estimates expected positive gain:

$$\hat{g}_i = \hat{p}_i \cdot \hat{g}_i^+ \approx E[g_i \mathbf{1}(g_i > 0) | x_i]. \quad (3)$$

This estimator excludes negative-gain magnitudes and therefore differs from net expected gain. Rewriting features use the current query and preceding history; reranking features additionally use the final query and its candidate statistics. True gains and relevance labels provide training supervision, not predictor inputs. The complete feature specification is given in Section IV.C.

D. Cost-Aware Selective Decision

Query rewriting typically requires a generative-model call, whereas reranking scores multiple query–document pairs. We normalize these costs into an additive form so that both operations can be compared under a shared budget. For budget ratio ρ , CASER applies threshold $\lambda(\rho)$ to their expected positive gains per unit cost:

$$s_i^w = \frac{\hat{g}_i^w}{c_w}, \quad s_i^r(a_i^w) = \frac{\hat{g}_i^r(a_i^w)}{c_r}. \quad (4)$$

An operation is executed when its unit-cost gain exceeds $\lambda(\rho)$ and skipped otherwise. Equivalently, the triggering conditions are:

$$\hat{g}_i^w - \lambda(\rho)c_w > 0, \quad \hat{g}_i^r(a_i^w) - \lambda(\rho)c_r > 0. \quad (5)$$

The shared threshold allows rewriting and reranking to compete for enhancement resources without predefined module budgets. However, reranking gain depends on the rewriting decision, operation costs are discrete, and the positive-gain estimator excludes negative-gain magnitudes. Consequently, this sequential policy is a cost-aware approximation, not a globally optimal solution to Eq. (1).

In offline evaluation, threshold search performs 60 bisection steps over $[0, 1]$, recomputing the sequential decisions and retaining the feasible candidate closest to B_ρ . The search uses predicted positive gains and operation costs only, without true gains, relevance labels, or evaluation quality metrics. Because threshold changes can alter the retrieval path, total cost need not be globally monotone; the search is therefore heuristic. Explicit cost checks ensure that the retained candidate satisfies the modeled budget, but do not guarantee the best allocation among all feasible policies.

IV. EXPERIMENTAL SETUP

A. Dataset

We conduct experiments on QReCC [2], an open-domain conversational question answering dataset containing dialogue histories, current questions, human rewrites, answers, and passage retrieval annotations. These passage annotations support evaluating both retrieval and reranking, rather than rewriting alone. We retain samples with valid passage annotations to construct gain predictor training and evaluation data.

When rewriting is skipped, retrieval uses the original current question; otherwise, it uses the human rewrite provided by QReCC. This setting isolates the rewriting-selection decision from variability in automatically generated rewrites. Our evaluation therefore examines enhancement selection under budget constraints, rather than rewriting generation quality; effectiveness with an automatic rewriter remains to be evaluated.

The rewriting training, development, and evaluation sets contain 5,798, 1,254, and 1,230 turns, respectively. Reranking training contains 11,596 path examples, covering Direct and Rewrite for each training turn. Joint evaluation aligns 1,230 turns by ID across both predictors and retrieval paths. These turns are labelled test in the rewriting records and dev in the reranking records, with no overlap with either predictor’s training IDs.

B. Retrieval, Rewriting, and Reranking Settings

BM25 retrieves the top-100 candidate documents from the QReCC passage corpus [10], and retrieval quality is evaluated against passage relevance annotations. When rewriting is selected, the provided human rewrite replaces the original query, keeping the selection decision separate from automatic generation quality.

The ms-marco-MiniLM-L6-v2 cross-encoder scores the retrieved candidates and orders them by relevance [19]. It reranks all 100 candidates with batch size 32 and maximum input length 512 tokens. Reranking gain is measured by the change in nDCG@10. Because rewriting can change the candidate set, gain prediction uses statistics from the actual retrieval path: Rewrite when rewriting is selected, and Direct otherwise.

C. Gain Predictor Training Settings

We train separate rewriting and reranking gain predictors using scikit-learn [20]. Each combines HistGradientBoostingClassifier for positive-gain probability with HistGradientBoostingRegressor for conditional positive-gain magnitude. Their product gives expected positive gain. The reranking predictor is trained jointly on Direct and Rewrite examples.

Both predictors use all preceding utterances in QReCC’s Context field, without future turns. Rewriting text input is `context [SEP] question`. Its numeric features are question and history token lengths; history utterance count; turn number; pronoun count and presence; comparison-word count and presence; first-question-word ID; capitalized-span count; quotation and digit presence; and token-set Jaccard similarities between the question and either the full history or the previous utterance. No retrieval information is used.

Reranking text input combines history, question, available rewrite, final query, and query-variant label. BM25 features comprise scores at ranks 1, 2, 5, and 10; means over the top 5, 10, and 100 candidates; population standard deviation over candidates; and gaps $s_1 - s_j$ and relative gaps $(s_1 - s_j) / \max(|s_1|, 10^{-9})$ for $j \in \{2, 5, 10\}$. Other features are query, question, history, and rewrite token and character

lengths; query pronoun count and presence; query question-word ID; query-history and question-rewrite Jaccard similarities; rewrite-path, query-equals-question, and query-equals-rewrite indicators; turn number; history utterance count; and reranker input candidate count. Candidate statistics follow the selected retrieval path; on Direct, rewrite text is empty and rewrite-dependent features are zero.

Text inputs use lowercase unigram/bigram TF-IDF with sublinear term frequency and 256-dimensional truncated SVD, concatenated with median-imputed and standardized numeric features.

D. Cost Settings and Budget Ratios

We count only additional enhancement costs relative to Direct, excluding base retrieval. Experiments use normalized operation costs $c_w = 1$ and $c_r = 2.12$. Retrieval quality is evaluated with QReCC human rewrites, while automatic rewriting latency is measured separately. The available timing records do not verify 2.12 as an empirical latency ratio, so it is treated as a fixed cost-model parameter.

The timing setup uses gpt-5.4-2026-03-05 for rewriting and ms-marco-MiniLM-L6-v2 for reranking 100 candidates, with batch size 32 and maximum input length 512 tokens. Wall-clock timing covers the API request-to-response interval for rewriting and scoring all candidates for reranking; five reranking warm-up queries are excluded.

Constant operation costs approximate latency that varies with text length, hardware, and inference settings. We report results at 10%, 30%, 50%, 70%, 90%, and 100% budget ratios and plot cost-quality curves. These results describe modeled enhancement cost rather than measured end-to-end latency savings.

E. Baselines and Evaluation Metrics

We compare CASER with Direct, Full Enhancement, and four fixed random allocation baselines. Direct performs neither rewriting nor reranking, while Full Enhancement applies both operations to every sample. CASER selects operations using expected positive gain per unit cost and the budget threshold.

The random baselines use no gain predictors. Rewrite First randomly selects samples for rewriting, then spends the remaining budget on random reranking; Rerank First reverses this order. Equal Cost divides the budget equally between the two modules and randomly selects samples within each pool, without transferring unused capacity. Equal Count randomly selects equal numbers of rewriting and reranking operations within the budget. Ablations replace gain-guided rewriting or reranking with random selection, or replace shared allocation with independent budgets, to assess each component’s contribution.

We use nDCG@10 as the primary retrieval-quality metric [21], with Recall@10, MRR@10, and Hit@10 as auxiliary metrics. Normalized enhancement cost and budget ratio measure modeled resource use. Main results compare nDCG@10 across budgets, while ablations report performance drops relative to CASER.

TABLE I
RETRIEVAL PERFORMANCE OF DIFFERENT METHODS UNDER DIFFERENT
BUDGET RATIOS MEASURED BY nDCG@10

Method	Budget Ratio					
	10%	30%	50%	70%	90%	100%
Rewrite First	0.1922	0.4449	0.5176	0.5679	0.6199	0.6457
Rerank First	0.0736	0.0838	0.0949	0.1386	0.4763	0.6457
Equal Cost	0.1345	0.2783	0.4339	0.5607	0.5865	0.6000
Equal Count	0.1127	0.2108	0.3232	0.4432	0.5758	0.6457
CASER	0.2296	0.4706	0.5517	0.6159	0.6435	0.6457

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Main Results

Table I reports nDCG@10 under different budget ratios. CASER outperforms the four fixed random allocation strategies at 10%, 30%, 50%, 70%, and 90% budgets, supporting selection based on predicted gains and operation costs rather than predefined allocation rules.

The reference baselines omitted from Table I are Direct and Full Enhancement, which achieve nDCG@10 of 0.0680 and 0.6457, respectively.

The 70% budget illustrates CASER’s cost–quality trade-off. CASER achieves nDCG@10 of 0.6159, retaining 95.39% of full-enhancement quality at approximately 70% of its modeled enhancement cost. Rewrite First, Equal Cost, and Equal Count obtain 0.5679, 0.5607, and 0.4432, respectively. These comparisons suggest that gain-guided selection allocates limited resources more effectively than the evaluated random strategies.

At 90% budget, CASER reaches 0.6435, close to full enhancement at 0.6457, suggesting diminishing benefits from additional operations. A 100% allowance does not necessarily produce full enhancement under every allocation rule. In particular, Equal Cost retains separate module budgets without redistributing unused capacity, so it does not apply both operations to every sample and reaches only 0.6000.

B. Cost–Quality Curve Analysis

Fig. 2 shows how retrieval effectiveness changes with the enhancement budget across different strategies.

CASER’s nDCG@10 rises rapidly at lower budgets and gradually stabilizes with slight fluctuations at higher budgets. This pattern is consistent with prioritizing operations with higher predicted gain per unit cost, followed by diminishing returns as additional operations are selected.

CASER’s advantage over random allocation is pronounced at 10%–70% budgets. Rerank First’s weak low-budget performance further motivates selecting reranking operations according to their expected benefit.

The dense budget sweep peaks at nDCG@10 of 0.6461 near 95.7% budget, slightly above full enhancement at 0.6457. Selectively skipping some operations can therefore match full-enhancement quality in this evaluation. However, without uncertainty estimates, the small difference does not establish a

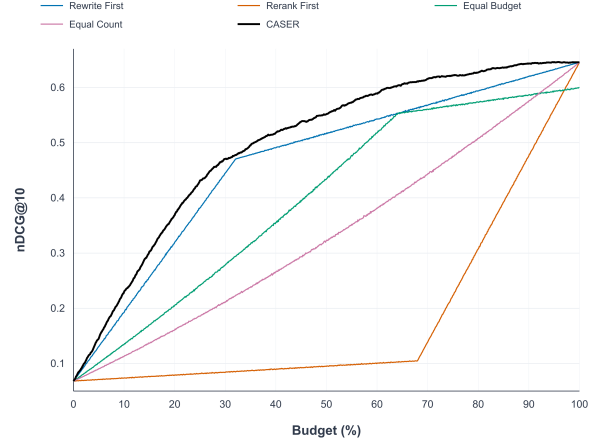


Fig. 2. Cost–quality curves under different budget ratios.

significant improvement. The main finding is near-full retrieval quality at lower modeled enhancement cost.

C. Ablation Study

We evaluate three ablation categories: replacing gain-guided rewriting with random selection, replacing gain-guided reranking with random selection, and replacing shared allocation with independent module budgets. The last category includes Equal Cost and Equal Count, yielding four variants in total.

Fig. 3 reports performance drops relative to CASER in nDCG@10, MRR@10, and Recall@10.

The variants generally degrade performance across most budget ranges, supporting both gain predictors and shared allocation. Random rewriting can spend resources on low-gain reformulations, while random reranking disregards differences in candidate states. Independent budgets restrict resource transfer between modules, whereas CASER allows both operations to compete under the same gain-per-cost threshold. These results support adaptive selection and allocation, without implying that every component improves every budget setting.

The plotted ablations retain both operations. Existing route results for the same 1,230 evaluation turns also provide two single-module operating points: rewriting all turns without reranking yields nDCG@10 of 0.4706 at $1/3.12 = 32.05\%$ of full enhancement cost; reranking all original-query results without rewriting yields 0.1045 at $2.12/3.12 = 67.95\%$. These results partially address the single-module question, but their costs differ. Matched-budget selective comparisons and experiments allowing both operations while charging only one cost remain unreported.

VI. CONCLUSION

We propose CASER to allocate a shared budget between optional query rewriting and document reranking using two-stage positive-gain prediction. Its sequential policy respects information availability but does not guarantee global optimality.

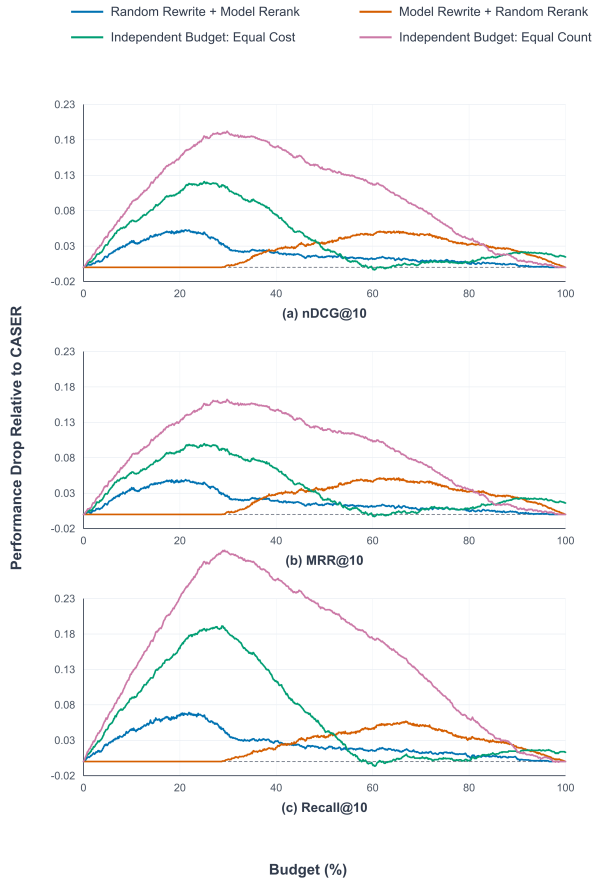


Fig. 3. Performance drop of ablation strategies relative to CASER.

On QReCC, CASER outperforms the evaluated random allocation strategies. At 70% budget, it achieves nDCG@10 of 0.6159, retaining 95.39% of full-enhancement quality. Ablations support both gain predictors and shared allocation in most budget ranges.

Evaluation is limited to QReCC, human rewrites, and a fixed cost ratio without verified latency calibration. Future work should examine other datasets and retrievers, generated rewrites, variable costs, matched-budget single-module and related-work comparisons, and online budget control.

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